Problem FoL.1 ... inconspicuous escalator

3 point

Suppose that the stairs of an escalator are moving with speed $v = 4.0\,\mathrm{km\cdot h^{-1}}$. How much ca
he speed of the moving belt the passengers hold onto differ from this speed (in percent), suc
hat the passengers cannot tell the difference between the two speeds? The escalator is $d = 30\,\mathrm{m}$
ong and a passengers cannot tell the difference if their hand moves by less than $\Delta x = 5\,\mathrm{cm}$
luring the whole journey from one end of the escalator to the other end.

Karel was inspired by Dodo's model, which he then disproved with this problem

Problem FoL.2 ... long day

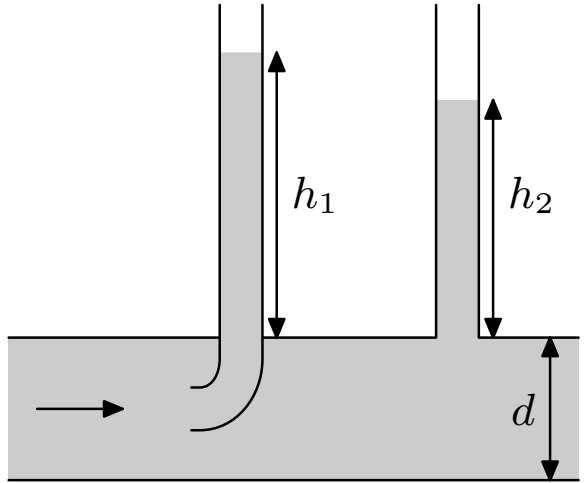
3 point

Consider a boat floating on the equator. How fast (in knots) does it have to move if one da
on the ship should last 25 hours? *A 24-hour day is too short for Matěj*

Problem FoL.3 ... pair of pipes

3 point

Water flows without friction in a horizontal pipe with a diameter $d = 10\,\mathrm{cm}$. The pipe is connected to two smaller vertical pipes. The intake of the first pipe faces against the water flow, so it is perpendicular to the horizontal pipe, and the water flowing into it comes to a stop in this pipe. The intake of the second pipe is parallel to the horizontal pipe. The water reaches a height $h_1 = 30.0\,\mathrm{cm}$ in the first smaller pipe, but it only reaches a height $h_2 = 25.0\,\mathrm{cm}$ in the second pipe. What is the velocity of the water flowing in the main pipe?



Jindra was dreaming of water in the summer dry season

Problem FoL.4 ... lost heat

3 point

A calorimeter contains 0.50 kg *of water at a temperature* $t_1 = 20\,^\circ\mathrm{C}$. *We add another* 0.50 kg *o water at a temperature* $t_2 = 80\,^\circ\mathrm{C}$ *to the calorimeter. After it has reached thermal equilibrium he water has temperature* $t_3 = 45\,^\circ\mathrm{C}$. *Determine the heat capacity of the calorimeter. Assum hat the calorimeter does not exchange heat with its surroundings.*

Jindra was curious where the heat from water in a calorimeter has gone to

Problem FoL.5 ... one hundred

3 point

Jindra wants to know the amount of heat that needs to be supplied to a cube of ice wit mass $1.0\,\mathrm{kg}$ *at the temperature* $0.0\,^{\circ}\mathrm{C}$ *in order to change it into steam at* $100.0\,^{\circ}\mathrm{C}$*. The specifi heat of water is* $1\,\mathrm{kcal}\cdot\mathrm{kg}^{-1}\cdot\mathrm{K}^{-1}$*; assume that it does not depend on temperature. Look up an other necessary constants.*

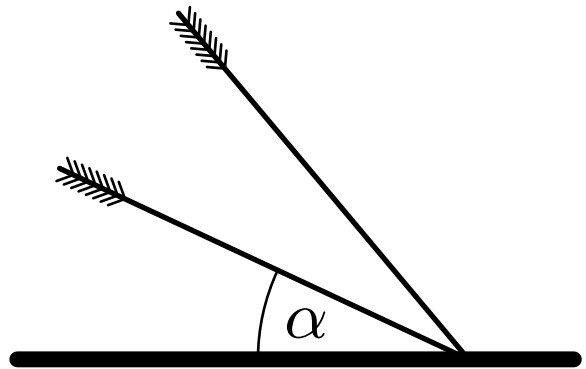
Jindra ordered boiled ice in the restauran

Problem FoL.6 ... archer

4 point

An archer is shooting arrows from a point on a horizontal plane. He manages to land two arrows at essentially the same points, but pointing in different directions. One of the arrows is pointing at an angle $\alpha = 25.00^\circ$ with respect to the horizontal plane. At what angle with respect to the horizontal plane is the second arrow pointing if we assume that both were launched with the same force?

Matěj was shooting arrows.



Problem FoL.7 ... strange eclipse

4 point

The maximum duration of a total solar eclipse on the Earth's surface is $t = 449\,\mathrm{s}$. *During thi ime, the stripe of land covered by the total eclipse (the shadow of the Sun) has length* $d =$ $= 267\,\mathrm{km}$. *Find out the maximum duration of a total solar eclipse for a passenger on a plan which flies with a velocity* $v = 903\,\mathrm{km/h}$. *Neglect the height of the plane above the surface.*

Dodo is dreaming about a trip to Argentina

Problem FoL.8 ... neglect causality

4 point

Elisa had a daughter Lottie with Johann. Unfortunately, something strange happened to Lottie She travelled back in time to the past, where she grew up and married Peter. She and Peter ha a daughter Elisa, which is also her mother. Therefore, a time paradox is created. The existenc of Lottie is caused by the existence of Elisa and vice versa, we can't claim that their existenc has a beginning. Assume that each child inherits exactly half of her genetic information from th father and half from the mother. What percentage of Elizabeth's genetic information originall came from Peter? Johann is not related to Peter.

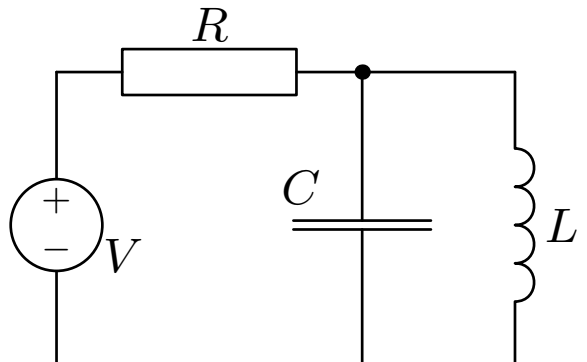
Matěj was watching a series, but I won't reveal its name to avoid spoilers

Problem FoL.9 ... fun with RLC

4 point

Consider the circuit in the figure. It consists of a coil with inductance $L = 10.0\,\mathrm{mH}$, *a capacitor with capacitance* $C = 4.70\,\mu\mathrm{F}$, *a resistor with resistance* $R = 1.00\,\mathrm{k\Omega}$ *and an ideal source of DC voltage* $V = 230\,\mathrm{V}$. *Calculate the real electric power consumed by the resistor. The utility frequency in Czech sockets is* $f = 50\,\mathrm{Hz}$.

Dodo wants to be cruel.



Problem FoL.10 ... an unfair race

4 point

Martin and Ivan had a race to the opposite side of a hill. Both of them ran the same horizonta
distance $x = 60\,\mathrm{m}$. *However, Ivan ran over the hill, which can be described as an arc of a circl*
with radius $r = 50\,\mathrm{m}$, *while Martin ran along a corridor through the hill. The floor of thi*
corridor is an inclined plane and its angle of inclination with respect to the horizontal plan
is $\alpha = 14\,^\circ$. *How much larger (in percent of Martin's average speed) must Ivan's average speed b*
if both runners reached the end of the corridor (on the opposite side of the hill) simultaneously

Martin did not feel like running up a hil

Problem FoL.11 ... Humber Bridge

3 point

The Humber Bridge, which runs across the river Humber in England, is one of the larges suspension bridges in the world. It consists of two 155.5 m *high towers (with height measure from the river surface), whose bases are* 1410 m *away from each other (Euclidean distance, no distance on the ground), connected by ropes that are holding a road. Although both towers ar vertical, their highest points are further away from each other than their bases. What is th difference of these (Euclidean) distances?*

Matěj was watching Tom Scot

Problem FoL.12 ... jumping

4 point

Find the angular velocity necessary to land a qaudruple Axel jump. The figure skater takes o
rom the left forward outside edge (the outer side of the left-leg ice skate's edge while movin
orward) and lands on the right back outside edge, while achieving a maximum height $h = 1.0\,\mathrm{m}$
bove the ice.

Dodo and the Grand Prix of Figure Skating

Problem FoL.13 ... spider-spider

4 point

Pete the spider is 2 cm *long and he can weave threads of spider web with different thickness but he needs to weave a thread whose radius is at least* 50.0 nm *in order for it to be able t support his own weight. What is the minimum radius of a thread that could support the spide if his size increased to that of a human? Assume that he would be* 2 m *long, with all his othe dimensions increasing accordingly, but the density remaining constant.*

Tomáš was learning mathematics

Problem FoL.14 ... is it colonizable?

4 point

A planet similar to the Earth has a mass $M = 7.166 \cdot 10^{24}\,\mathrm{kg}$*, which is a bit more than th Earth's mass. By coincidence, the first and second cosmic velocities on this planet are the sam s on the Earth. What is the gravitational acceleration on its surface?*

Matej can't wait for colonization

Problem FoL.15 ... Dan's fléche burnt.

5 point

Jáchym spilled some lava in Minecraft. Let's assume (in order to simplify) that the world i a 2D cube grid with edge length 1 m. *We can simulate the lava by the following set of rules:*

1. *the lava level in the central cube is* 1 m,
2. *the lava level in any block which cannot be reached from the central block by movin through less than 3 other cubes (assume that we can only move between cubes if the have a common face), is* 0,
3. *the lava level in other blocks is given by the arithmetic mean of levels in neighbourin blocks (two blocks are neighbours if they have a common face).*

What is the total volume of the spilled lava?

Jáchym poured a bucket of lava on Dan's fléche in Minecraf

Problem FoL.16 ... one hundred reloaded

4 point

Jindra was interested in the magnitude of error in the problem "one hundred". The specifi heat of water is in fact temperature-dependent. Jindra found an empirical relationship for th specific heat $c_{\mathrm{water}} = 3.1832 \cdot 10^{-6} t^4 - 7.7922 \cdot 10^{-4} t^3 + 7.5387 \cdot 10^{-2} t^2 - 2.9190 t + 4.2158 \cdot 10$ *where the temperature* t *must be in* °C *and the specific heat is obtained in* $\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$. *What i the difference (in percent) between the heat supplied to water with a constant specific heat* $c =$ $= 4184.0\,\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$ *and the heat calculated using the empirical relationship, if the mass o this water is* $m = 1.00\,\mathrm{kg}$ *and its temperature increases from* $0.00\,^{\circ}\mathrm{C}$ *to* $100.00\,^{\circ}\mathrm{C}$ *in both cases The sign of the result is important (i.e. if the amount of heat for the case with constant specifi heat is lower, we expect a negative sign).*

Jindra received steam instead of boiled ice

Problem FoL.17 ... lazy electron

4 point

Consider the Bohr model of a hydrogen atom – a positively charged nucleus with charge e = $= 1.6 \cdot 10^{-19}$ C *and an orbiting electron with the opposite charge and a much smaller mass* m_e = $= 9.1 \cdot 10^{-31}$ kg. *What would be the radius of the hydrogen atom (in kilometers) if the perio of one orbit of the electron took one day? Do not consider the fact that the hydrogen aton could not exist in such a state (or could it...?).*

Matěj was comparing a hydrogen atom with the globe

Problem FoL.18 ... charge in a box

4 point

A nasty charge has escaped from an imaginary cylinder, so Jindra locked it into a box. The bo
has dimensions $8.00 \times 8.00 \times 4.00\,\mathrm{cm}$. *Jindra taped the charge into the middle of its square base*
The magnitude of the charge is $Q = 1.00 \cdot 10^{-6}\,\mathrm{C}$. *Find the electric flux through the opposit*
face of the box. *However, electric charges don't like Jindra*

Problem FoL.19 ... helix

4 point

A positron is moving at a distance $R = 10\,\mathrm{cm}$ *from a wire carrying a current* $I = 2.5\,\mathrm{A}$*. Th current generates a magnetic field, which causes the positron to orbit around the wire. Th component of the positron's velocity in the direction of the current is* $v = 1.5\,\mathrm{m{\cdot}s^{-1}}$ *and th adial component is zero. Under the condition that the movement is stable, determine the perio of the positron's orbit around the wire (in miliseconds).*

Let $e = 1.6 \cdot 10^{-19}\,\mathrm{C}$, $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{N{\cdot}A^{-2}}$, $m_\mathrm{e} = 9.1 \cdot 10^{-31}\,\mathrm{kg}$. *You spin me right round*

Problem FoL.20 ... charge inside a cylinder

6 point

n a thought experiment, Jindra managed to catch an electric charge $Q = 1.00 \cdot 10^{-6}\,\mathrm{C}$ *int*
cylinder which does not interact with the charge. The charge remains stationary in the centr
f the cylinder. The cylinder has radius $r = 4.00\,\mathrm{cm}$ *and height* $v = 6.00\,\mathrm{cm}$*. Determine th*
lectric flux through the top base of the cylinder. *Jindra likes electric charges*

Problem FoL.21 ... African Sun

4 point

Mišo went on another train adventure – he wanted to cross the entire Africa by train. In th middle of September, before arriving to Nairobi, he saw a reflection of the just-rising Sun i he window on the right while sitting in the direction of the route (facing in this direction). A hat time, the train was moving exactly south. 10 minutes later, the train was moving south outhwest. Determine the angle (in degrees) by which the reflection of the Sun with respect t he wagon deviated from the original position from Mišo's point of view.

Dodo was going home by a night train

Problem FoL.22 ... Red hot nickel... cable?

4 point

A heater has a nickel heating filament with a temperature coefficient of resistance $\alpha = 0.0068\,\mathrm{K}^{-}$ *and with a resistance* $R_0 = 80\,\Omega$ *at the room temperature* $T_0 = 293\,\mathrm{K}$. *After turning on th heater, the filament becomes red-hot at the temperature* $T = 1100\,\mathrm{K}$ *thanks to a current* $I =$ $= 2\,\mathrm{A}$ *passing through it. What is the total surface of the filament (in* m^2*), assuming that th filament transfers heat only by radiation (acts as a black body)?*

Jirka is already thinking about how to warm up in winte

Problem FoL.23 ... hanging here

5 point

Two springs with identical length $l = 5.0\,\mathrm{m}$ *and stiffness* $k = 1.0.10^5\,\mathrm{N{\cdot}m^{-1}}$ *are stretche horizontally between two vertical walls (at the distance* $2l$*) in such a way that the springs ar connected in the middle. Next, we hang a mass* $m = 3.0\,\mathrm{kg}$ *at the point where the springs ar connected. What is the height* h *by which the point of connection drops? You may assume tha* $h \ll l$ *and* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

The tension is rising

Problem FoL.24 ... doggy

6 point

Suppose that we are on a hypothetical planet Caniculium orbiting the white dwarf Sirius B which is the weaker component of the Sirius binary system. From the orbit of this planet a hybrid eclipse of Sirius A can be observed (in a hybrid eclipse, the stars have the same angular diameter). Find out the proportion of landscape illumination (power incident on the surface of the planet) during the eclipse compared with the situation just before the eclipse starts. The effective surface temperatures of these stars are $T_A = 9940\,\mathrm{K}$ $T_B = 25\,000\mathrm{K}$.

Dodo thinks that the Moon is not shining by itself

Problem FoL.25 ... ethanol emissions

3 point

We want to completely burn $V_e = 1.00\,\mathrm{dm}^3$ *of liquid ethanol in an oxygen rich atmosphere, s*
hat the main product of the combustion is carbon dioxide (CO_2)*. All the released carbon diox*
de should be contained in a vessel at the standard atmospheric pressure and temperature 0°C
What is the required volume of such a vessel? (Assume that we can eliminate all other reactio
products.)

Karel was thinking about experiments

Problem FoL.26 ... jump into the unknown

4 point

A very thin cylinder with a length $l = 32.7\,\mathrm{cm}$ is in a vertical position with its bottom bas at the height $10\,l$ above a water surface. What depth below the water surface does the lowes point of the cylinder reach after we let it fall? The density of the liquid is three times highe than the density of the cylinder. Neglect any drag, friction and surface tension.

Jáchym fell into the water in Minecraf

Problem FoL.27 ... rolling cone

5 point

On a horizontal surface, a cone with a base radius $r = 10.0\,\mathrm{cm}$ *and height* $H = 100.0\,\mathrm{cm}$ *mad*
rom a material with density $\varrho = 1\,253\,\mathrm{kg{\cdot}m^{-3}}$ *is slipping with negligible rolling friction. Th*
ngular speed of rotation of the cone around the axis perpendicular to the surface and passin
hrough its apex is $\omega = 2.50\,\mathrm{rad{\cdot}s^{-1}}$*. Determine the magnitude of the force which the con*
xerts on the surface.

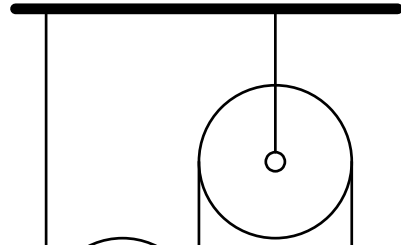
Dodo's head was spinning

Problem FoL.28 ... pulleys

5 point

Find the acceleration of the lower pulley and its direction (the result should be negative if the pulley accelerates downwards or positive if it accelerates upwards). Neglect moments of inertia of the pulleys.

Matěj wanted to invent an infinite system of pulleys, but it didn't work.

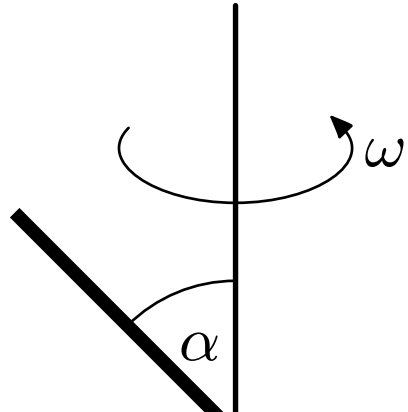


Problem FoL.29 ... small torque

5 point

A thin homogeneous rod is attached to an axis of rotation passing through the rod's centre. The angle between the rod and the axis of rotation is $\alpha = 45°$. *The rod has mass* $m = 40\,\mathrm{g}$, *length* $l = 30\,\mathrm{cm}$ *and it rotates with an angular velocity* $\omega = 25\,\mathrm{rad{\cdot}s^{-1}}$. *What is the magnitude of the torque by which the rod acts on its point of contact with the axis?*

Vindra's problem with the inertia tensor didn't pass, so he invented this.



Problem FoL.30 ... asteroid with impact parameter

5 point

An asteroid very, very far from the Sun (we may assume it is infinitely far) has a speed $v = 3.9\,\mathrm{km\cdot s^{-1}}$. *If there was no gravitational influence on the asteroid, the smallest distanc between the asteroid and the Sun would be* 1 au. *What will be the smallest distance betwee the asteroid and the Sun when the Sun's gravitational force is acting on the asteroid?*

Karel was thinking about alien spacecrafts

Problem FoL.31 ... defective spherical mirror

5 point

An astronomer wants to photograph a star using a Newtonian telescope. The primary mirror o the telescope is spherical (not parabolic) with a radius of curvature $R = 1.00\,\mathrm{m}$ *and transvers diameter* $d = 10.0\,\mathrm{cm}$*. The astronomer placed a CCD sensor in the focal plane at a distance* $R/$ *from the mirror's vertex. What will be the radius of the star's image on the chip? Assume tha the star is a point source of light.* *Jindra was thinking about imperfections of the world. .*

Problem FoL.32 ... quantum Kepler

5 point

As you surely know, Kepler's third law is the relationship between the semi-major axis and th orbital period of celestial bodies $a^3/T^2 = \text{const}$. *This relationship can be also derived from th nature of the gravitational field. The electric field is also inversely proportional to the squar of distance, so the situation is analogous to the gravitational field. Derive how Kepler's thir law would look for the system of an electron and a proton if they behaved according to th classical laws of physics. What would be the ratio of the third power of the semi-major axi and the square of the orbital period?*

Jindra wondered if hydrogen orbitals could be explained using Platonic bodies

Problem FoL.33 ... tense tree

6 point

Jáchym cut down a homogeneous cylindrical tree with height $l = 30.0\,\mathrm{m}$. The trunk of the tre is still partially connected to the stump, so the top of the tree follows a circular trajectory Find the distance from the stump to the point where the normal stress in the tree is zero at th moment when the angle between the tree and the ground is $\alpha = 47°$. The connection betwee the trunk and the stump isn't causing any energy losses. *Jáchym fell(ed a tree)*

Problem FoL.34 ... slim line

5 point

Jindra is returning from a space mission and is moving towards the Earth with the speed v =
= 0.300 c. Danka, whose rest mass is 50.0 kg, *is looking forward to meeting Jindra after a lon*
ime, so she set off in a rocket towards Jindra's spaceship. However, Danka has been on a diet
She doesn't want Jindra to be able to measure her relativistic mass being greater than 60.0 kg
What maximal speed u with respect to the Earth can Danka fly? The answer is the rati
etween u and c. *For Jindra, travelling around Earth is not enough*

Problem FoL.35 ... overheated flywheel

4 point

By how many degrees Kelvin will a flywheel heat up by stopping? The initial frequency o rotation of the flywheel is $f = 120\,\mathrm{Hz}$. *Assume that half of its kinetic energy will be used fo heating it up. The flywheel is a homogeneous cylinder with a radius* $r = 15\,\mathrm{cm}$ *and it is made o steel with a specific heat capacity* $c = 450\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$. *Hint: we don't need to know any othe quantities, only the value of* π.

Karel was discussing IYPT tasks

Problem FoL.36 ... looping

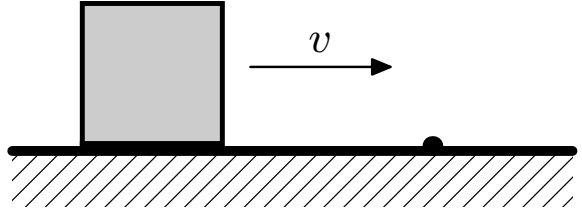
6 point

magine a biker who wants to ride around a thin vertical circular looping (a cylindrical surfac uch that its axis is horizontal) with a radius $R = 10.0\,\mathrm{m}$. Therefore, the looping has no entr r exit and the biker starts from the lowest point with a constant angular acceleration. What i ts minimum value such that the motorcycle can drive through the looping while always keepin n contact with the looping (without falling)? Use $g = 9.81\,\mathrm{kg{\cdot}s^{-2}}$.

Matěj has driven through a looping by car many times

Problem FoL.37 ... bumps on the road

6 point



A homogeneous cube with edge length $a = 1.0 \cdot 10^{-2}$ m is happily sliding with a velocity v on a horizontal plane without friction such that one face (hence two) is perpendicular to the direction of movement. Directly in front of it, there is a low stopper, and the cube collides with it perfectly inelastically; the point of impact is in the middle of its bottom front edge. Find the minimum velocity v such that the cube flips over the bump (stopper).

Matej calculated this during a college party

Problem FoL.38 ... speed merchant

4 point

Dan is driving on a straight road with a velocity $v_\mathrm{d} = 90\,\mathrm{km{\cdot}h^{-1}}$*. In the distance* $d = 110\,\mathrm{m}$ *in front of him, he notices a cyclist riding with a velocity* $v_\mathrm{c} = 18\,\mathrm{km{\cdot}h^{-1}}$*. Because there ar cars coming from the opposite direction, Dan can't drive around the cyclist, so he has to hi the brakes. His response time is* $t_\mathrm{r} = 0.50\,\mathrm{s}$*. What is the lowest allowed value of acceleratio (deceleration due to the brakes) of the car such that Dan doesn't crush the cyclist? The ca decelerates uniformly.*

Jindra rode with Dan in a car once and had enough

Problem FoL.39 ... slowly thrown rock

6 point

da is one of the few asteroids in the main asteroid belt that has its own natural satelite. Th moon's name is Dactyl. We haven't yet measured the shape of its orbit very precisely. Suppos hat its distance from Ida in the pericenter is $r_\mathrm{p} = 100\,\mathrm{km}$ *and in the apocenter is* $r_\mathrm{a} = 200\,\mathrm{km}$ *Dactyl's mass is* $m_\mathrm{D} = 10^{10}\,\mathrm{kg}$*, the mass of Ida is* $m_\mathrm{I} = 5 \cdot 10^{16}\,\mathrm{kg}$*. Imagine an astronaut wit mass* $m = 100\,\mathrm{kg}$ *on the surface of Dactyl. In the pericenter, the astronaut jumps up in th direction of motion of Dactyl with a velocity* $v = 5\,\mathrm{m{\cdot}s^{-1}}$*. Determine the displacement of th apocenter distance of Dactyl caused by the jump (a positive number means that this distanc ncreases). Neglect gravitational effects of the astronaut, i.e. treat it as a two-body problem.*

Mirek gains inspiration at conference lectures

Problem FoL.40 ... aesthetic balls

6 point

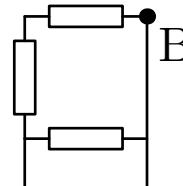
We have two small charged balls with the same mass hanging on strings (with negligible mass rom one point at the ceiling. Both balls have the same charge and they are in a homogeneou gravitational field. The angle between the strings is $10°$*. Then, we double the charge of one ba nd increase the charge of the other ball* k *times. Find the value of* k *which is necessary if w vant the angle between the strings to be* $25°$*.* *Karel heard about Danka's plans*

Problem FoL.41 ... disgusting resistive skyscraper

7 point

Determine the resistance between the points A and B if each resistor has resistance $R = 3141.59\,\Omega$.

Matej was reminiscing about Physics Olympiad preparation.



Problem FoL.42 ... manoeuvre

6 point

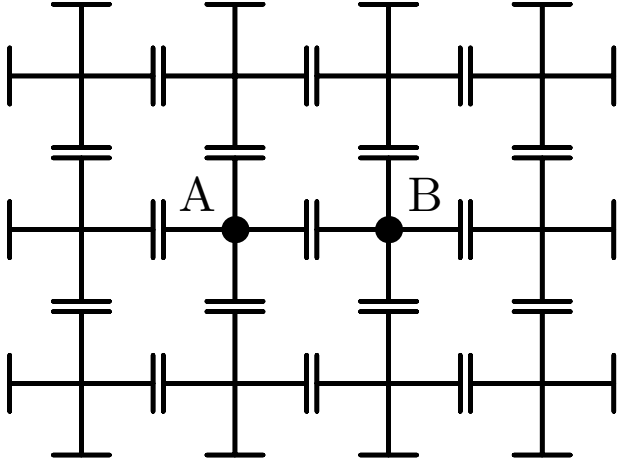
magine that we are operating a spacecraft on a circular orbit around a star and we wish t
louble the radius of our circular orbit. We choose to achieve this by a specific manoeuvre
ve maintain a constant velocity – with a constant magnitude and a constant angle $\theta = 5$
etween the velocity vector and the tangent line to the local circular orbit. How many orbit
not necessarily an integer) do we make before we reach the desired distance from the star?

Mirek came up with a problem which turned out to be too difficult for elementary schoo

Problem FoL.43 ... capacitor grid

6 point

Consider a square lattice made from capacitors. Part of the lattice is shown in the figure. Th capacitance of each capacitor is $C = 100.0\,\mu\mathrm{F}$. *Find the total capacitance between two adjacen vertices.*



Matěj would like to have an infinite grid at home

Problem FoL.44 ... the fourth velocity

5 point

The particle velocity distribution in a gas is described by the Maxwell-Boltzmann distributio

$$f(v) = \sqrt{\frac{2}{\pi}} \left(\frac{M}{RT}\right)^{3/2} v^2 \mathrm{e}^{-\frac{Mv^2}{2RT}} .$$

There are several ways to determine an average velocity. We define the most probable veloc
ty $v_\mathrm{p} = \sqrt{\frac{2RT}{M}}$*, the mean absolute velocity* $v_\mathrm{s} = \sqrt{\frac{8RT}{\pi M}}$ *and the mean quadratic velocity* $v_\mathrm{k} =$
$= \sqrt{\frac{3RT}{M}}$*. For the oxygen* O_2 *at temperature* $20\,^\circ\mathrm{C}$*, these velocities have values* $390\,\mathrm{m{\cdot}s^{-1}}$
$440\,\mathrm{m{\cdot}s^{-1}}$ *and* $478\,\mathrm{m{\cdot}s^{-1}}$ *respectively. However, we can also calculate the* n*-th moment of th*
Maxwell-Boltzmann distribution for any n *and thus find some other average speed. What i*
he value of the mean cubic absolute velocity of oxygen under the same conditions?

An oxygen molecule hit Jindra's eye

Problem FoL.45 ... inside a spherical mirror

6 point

Imagine two concave spherical mirrors with a common axis, such that the mirrors are two part of a sphere with a radius $r = 1\,\mathrm{m}$. When we place a point light source in the centre of th sphere, each mirror forms a reflected image of the source which coincides with the source itsel Find the distance of the source from the centre such that it coincides with its image forme after three reflections (but does not coincide with the image formed after one reflection).

Matěj was inspired by a Vsauce video

Problem FoL.46 ... rotary tree

5 point

Jachym cut down a tree that can be approximated by a thin homogeneous cylinder wit
ength $l = 30.0\,\mathrm{m}$. *The tree is still partly connected to the stump, so during its fall, th*
ip of the tree travels along a circular arc. There is a stone with negligible height lying on th
ground at a distance $l/9$ *away from the stump. Just before the tree hits the ground, it hit*
he rock in a perfectly inelastic manner. This breaks the connection between the tree and th
tump and the tree continues to move as a free body. With what speed does the tip of th
ree hit the ground? The connection to the stump or its breaking don't involve any losses i
nergy. *This year, Jáchym is going to get his Christmas tree from a forest*

Problem FoL.47 ... neutron rack

7 point

Find the smallest distance from the surface of a neutron star at which a free-falling steel rod ca exist, pointing towards the star, without being ripped in half. The star has mass $M = 1.80\,M_{\odot}$ *and radius* $R = 10.0\,\mathrm{km}$. *The ultimate tensile strength of the rod is* $\sigma = 800\,\mathrm{MPa}$, *its densit is* $\varrho = 7\,900\,\mathrm{kg{\cdot}m^{-3}}$ *and its length is* $L = 1.00\,\mathrm{m}$. *Neglect deformation of the rod which occu before fracturing and consider only Newtonian mechanics.*

Dodo was reading a sci-fi from 196

Problem FoL.48 ... sunbathing session

6 point

The average solar irradiance of the Earth (power incident on the Earth's surface) oscillate with a period of about 11 *years. A big part of the variance constitutes of changes in the UV part of the EM spectrum. Assume, for the purposes of this problem, that the Sun in the sola maximum is a blackbody with a temperature* $T = 5.800\,\mathrm{K}$. *In the solar minimum, the spectra characteristic is the same, except for a region of wavelengths between* 10 nm *and* 300 nm, *whic is missing. Determine the total drop in power emitted by the Sun between the maximum an minimum, in percent of the value in the maximum.*

Mirek gains inspiration at conference lectures

Problem FoL.49 ... diabolical spring

8 point

Let's consider a straight elastic rubber rod (i.e. non-circular rubber band) with rest length l =
= 9.57 cm, spring constant $k = 11.08\,\mathrm{N{\cdot}m^{-1}}$, mass $m = 24.53\,\mathrm{g}$ and cross-sectional area S =
= 5.70 cm^2. One end of the rod is connected with a pad underneath it (not vertically glued t
he pad, i.e. the only constraint is that this end cannot move away from the pad). Now, w
ill the space around the rubber rod with water; the density of water is $\varrho = 998\,\mathrm{kg{\cdot}m^{-3}}$. Wha
will the new equilibrium length of the rod be? Assume that the cross-sectional area of the ro
always remains constant.

You thought the last year's evil spring problem was hard? I will show you hard... – Jáchym

Problem FoL.50 ... the charge and the pentagram

9 point

Jindra has had enough with the free charges so he decided to try some magic. He took a regula pentagon with side length 1 *and cut out a pentagram. Then he caught a free point charge* $Q =$ $= 1.00 \cdot 10^{-6}$ C *and placed it directly above the centre of the pentagram at a distance* $\frac{1}{2}\sqrt{\frac{5-\sqrt{5}}{10}}$ *What is the total electric flux through the pentagram? Jindra still doesn't give up..*

Problem M.1 ... Rick's microplanet

3 point

Imagine that you are hiding from the Galactic Federation on a planet which has the sam gravitational acceleration on the equator as the Earth ($a_g = 9.83\,\mathrm{m\cdot s^{-2}}$), as well as the sam centrifugal acceleration on the equator ($a_c = 0.034\,\mathrm{m\cdot s^{-2}}$), but its radius is only $R = 5\,\mathrm{km}$ How long would one day on this planet be?

Consider the planet to be an ideal homogeneous sphere. The period of the planet's orbi around its star is negligible compared to the length of the day.

Karel and Matěj were wondering where Rick and Morty's family escaped

Problem M.2 ... interdimensional density

3 point

magine that you are still hiding on the planet from the previous problem. This planet ha he same gravitational acceleration on the equator ($a_g = 9.83\,\mathrm{ms}^{-2}$) and the same centrifuga cceleration ($a_c = 0.034\,\mathrm{ms}^{-2}$) as the Earth, but its radius is only $R = 5\,\mathrm{km}$. What is th lensity of this planet?

Assume that the planet is an ideal homogeneous sphere.

Karel was watching where Rick and Morty's family escaped

Problem M.3 ... squanchy frisbee

4 point

Consider the planet from the previous problems. This planet has the same gravitational acceler
ation on the equator ($a_g = 9.83\,\mathrm{ms}^{-2}$) and the same centrifugal acceleration ($a_c = 0.034\,\mathrm{ms}^{-2}$
as the Earth, but its radius is only $R = 5\,\mathrm{km}$. Life on the planet is boring, so you throw a
object eastward along the equator. You want it to travel around the planet and return back t
you from the west. How long does it take the object to come back?

Assume that the planet is an ideal homogeneous sphere and air drag is negligible. On thi
planet, the stars disappear on the western horizon. *Matej had nobody to play with*

Problem M.4 ... wubba lubba dub dub

4 point

We'll stay on the planet from the previous problems. This planet has the same gravitationa cceleration on the equator ($a_g = 9.83\,\mathrm{ms}^{-2}$) and the same centrifugal acceleration (a_c = = $0.034\,\mathrm{ms}^{-2}$) as the Earth, but its radius is only $R = 5\,\mathrm{km}$. How high above the surface woul he orbit of a geostationary satellite be, if it existed at all? If it does not exist, fill in 0 as th esult.

Assume that the planet is an ideal homogeneous sphere.

Karel was watching where Rick and Morty's family escaped

Problem E.1 ... breakers

3 point

The FYKOS-bird has two cylindrical copper wires, each with radius $r = 1\,\mathrm{mm}$ *and resistiv*
ty $\varrho = 1.69 \cdot 10^{-8}\,\Omega\cdot\mathrm{m}$. *He wants to make an extension cord, but he connects two ends o*
ne wire to a socket by mistake, so the wire short-circuits the socket. What is the maximun
ength of the wire which activates the circuit breaker? The voltage in the socket is $230\,\mathrm{V}$ *an*
he maximum allowed current through the breaker is $16\,\mathrm{A}$.

Mišo short-circuited

Problem E.2 ... Fykonacci

3 point

The FYKOS-bird was playing with some resistors. He connected two points A and B by a resisto with resistance $R_1 = 1.600\,\Omega$. He thought it was too much, so he added another resistor wit he same resistance $R_2 = 1.600\,\Omega$ between these points, parallel to the first one. Yet, he wa still not satisfied, so he added a third resistor $R_3 = R_1 + R_2 = 3.200\,\Omega$, parallel to the others Then he went bonkers and started adding more and more resistors parallel to the previous ones with resistances $R_4, R_5, \ldots$ described by the formula $R_n = R_{n-2} + R_{n-1}$, where $n \in N$. Wha was the final resistance between points A and B?

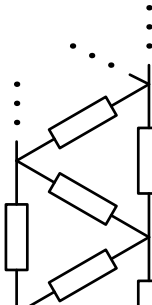
Jáchym does not enjoy physics, so he tries to create some math problems

Problem E.3 ... triangular ladder

4 point

The FYKOS-bird wanted to climb onto a certain resistive skyscraper which you might also encounter. Therefore, he built a triangular ladder (infinite, of course), made wholly of resistors with resistance $R = 1.0\,\Omega$. Calculate the resistance between the legs of the ladder.

Lego was infinitely sad because none of his other tasks made it through.



Problem E.4 ... nonlinear FYKOS-bird

4 point

The FYKOS-bird was tired of connecting resistors and so he decided to connect himself in series with two batteries, each with voltage $U_\mathrm{b} = 4.50\,\mathrm{V}$, *and a resistor with resistance* $R = 50.0\,\Omega$. *The bird behaves as a non-linear component with the current–voltage characteristic shown in the figure 4. The cut-in voltage of the bird is* $U_0 = 1.50\,\mathrm{V}$, *the saturation voltage of the bird is* $U_\mathrm{n} = 6.00\,\mathrm{V}$ *and the saturation current through the bird is* $I_\mathrm{n} = 250\,\mathrm{mA}$. *In each of the three parts of the current–voltage characteristic, the bird behaves as a linear component. Determine the current flowing through the bird.* *Dodo remembered labs at midnight.*

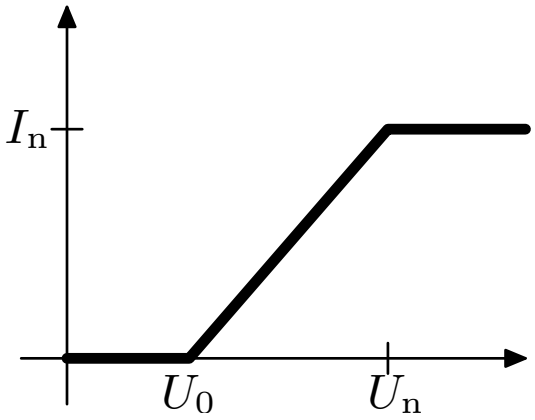
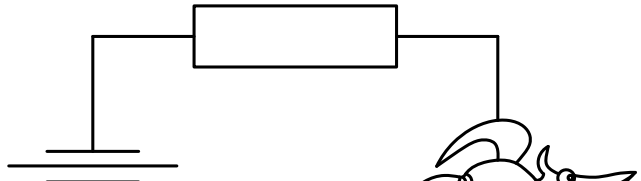


Fig. 4: I-V curve



Problem X.1 ... Chernobyl

3 point

Soon after the reactor no. 4 exploded, a radiation of 5 roentgens per hour was measured a a distance of 800 m *from the reactor. How much would have been measured at the distanc of* 200 m *from the reactor? Neglect the effects of air and environment on the propagation o radiation.*

Matěj was watching the series

Problem X.2 ... hold my graphite

3 point

An unsuspecting firefighter grabs a piece of graphite with an activity of $1.00 \cdot 10^{10}\,\mathrm{Bq}$*, lyin*
near the now defunct reactor. How many particles will go into (or through) his hand if th
firefighter holds the graphite for 10 seconds*? Consider the chunk of graphite to be a sphere wit*
a diameter of 5 cm *and assume that the firefighter's hand covers one fifth of its surface whe*
holding it. Assume further that each particle which leaves the piece of graphite corresponds t
the decay of one nucleus.

Matěj was chilling while watching meme

Problem X.3 ... not great, not terrible

3 point

Calculate the activity of a destroyed nuclear reactor. We measure 10000 radiation particles pe
minute per square decimeter at a distance of 5.00 km *from the reactor. Neglect the effects of ai*
and environment on the propagation of radiation. *Matěj wanted to set the answer to* 3.6 R

Problem X.4 ... radio-activity

3 point

By how many percent is the radiation around a destroyed nuclear power plant reduced afte
a 2.00m thick sarcophagus is built around it? Assume that 3.00 dm *of the material which th*
sarcophagus is made of can capture, on average, 50.0 % *of the radiation.*

Matěj is worried about his safety